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Application No.	Ref.	Date
20 794 947.0 - 1001	EUR1620WEP	28.07.2023
Applicant Ftlab Co., Ltd.		

Communication under Rule 71(3) EPC

1. Intention to grant

You are informed that the examining division intends to grant a European patent on the basis of the above application, with the text and drawings and the related bibliographic data as indicated below.

A copy of the relevant documents is enclosed.

1.1 In the text for the Contracting States:

AL AT BE BG CH CY CZ DE DK EE ES FI FR GB GR HR HU IE IS IT LI LT LU LV MC MK MT NL NO PL PT RO RS SE SI SK SM TR

Description, Pages

1-16	filed in electronic form on	20-04-2022

Claims, Numbers

1-15	filed in electronic form on	20-04-2022

Drawings, Sheets

1/8-8/8	filed with entry into the regional phase before the EPO

With the following amendments to the above-mentioned documents proposed by the division

Description, Pages	2, 3, 6-8, 11, 13, 15
Claims, Numbers	1 (amended), 10-11 (deleted), 12-17 (renumbered 10-15)

Comments

DESCRIPTION
Pages 2, 6-8, 13, 15: Invention scope - Rule 42(1)(c)
Pages 3, 6-8, 11, 15: Description adapted to amended claims (Art. 84 EPC)
Page 8: Inconsistency between claim and description removed (Art. 84 EPC)
CLAIMS
Page 1, Claim 1: Scope of claim unclear - clarified (Art. 84 EPC). The reference to an outdoor TV antenna shape was removed as there is no standardly defined shape for such an object.
Page 1, Claim 1: Absent or incorrect two-part form of claim (Rule 43(1) EPC)
Page 3, CLAIMS 10, 11: Inconsistency between claims removed (Art. 84 EPC). Original claims 10 and 11 were deleted as the features were already present in claim 1, and their inclusion created clarity issues with how the probe shape should be interpreted.
Pages 3-5, CLAIMS 12-15, 17: Claims renumbered to 10-13, 15 respectively.
Page 4, Claim 16: Claim renumbered to 14 and dependency corrected.

See also the comments in enclosed EPO Form 2906.

1.2 Bibliographic data

The title of the invention in the three official languages of the European Patent Office, the international patent classification, the designated contracting states, the registered name(s) of the applicant(s) and the other bibliographic data are shown on **EPO Form 2056** (enclosed).

2. Invitation

You are invited, **within a non-extendable period of four months** of notification of this communication,

2.1 to EITHER approve the text communicated above and verify the bibliographic data (Rule 71(5) EPC)

(1) by filing a translation of the claim(s) in the other two official languages of the EPO

		Fee code	EUR
(2a)	by paying the fee for grant including the fee for publication:	**007**	1040.00
	minus any amount already paid (Rule 71a(5) EPC):		0.00
		Total amount:	1040.00
(3)	by paying additional claims fees under Rule 71(4) EPC; number of claims fees payable: 0	**016**	0.00
	minus any amount already paid (Rule 71a(5) EPC):		0.00
		Total amount:	0.00

Important: If the translations of the claims and fees have already been filed and paid respectively in reply to a previous communication under Rule 71(3) EPC, e.g. in the case of resumption of examination after approval (see Guidelines C-V, 6), **agreement as to the text to be granted** (Rule 71a(1) EPC) must be expressed within the same time limit (e.g. by approving the text and verifying the bibliographic data, by confirming that grant proceedings can go ahead with the documents on file and/or by stating which translations of the claims already on file are to be used).

Note 1: See "Notes concerning fee payments" below.
Note 2: Any overpaid "minus" amounts will be refunded when the decision to grant (EPO Form

2006A) has been issued.

Note 3: For the calculation of the grant fee under Article 2(2), No. 7, RFees (old fee structure), the number of pages is determined on the basis of a clean copy of the application documents, in which text deleted as a result of any amendments by the examining division is not shown. Such clean copy is made available via on-line file inspection only.

2.2 OR, in the case of disapproval, to request reasoned amendments or corrections to the text communicated above or keep to the latest text submitted by you (Rule 71(6) EPC).

In this case the translations of the claims and fee payments mentioned under point 2.1 above are NOT due.

The terms "amendment(s)" and "correction(s)" refer only to amendments or corrections of the application documents and not of other documents (e.g. bibliographic data, the designation of the inventor, etc.).

If filing amendments, you must identify them and indicate the basis for them in the application as filed. Failure to meet either requirement may lead to a communication from the examining division requesting that you correct this deficiency (Rule 137(4) EPC).

2.3 Bibliographic data

Where you request a change or correction of bibliographic data in response to the Rule 71(3) communication, this will **not** cause the sending of a further communication under Rule 71(3) EPC. You will still have to pay the fees and file translations in reply to the Rule 71(3) communication in the case of 2.1 above, unless you also file a reasoned request for amendments or corrections in response to the Rule 71(3) communication (see case 2.2 above).

3. Loss of rights

If neither of the two possible actions above (see points 2.1 or 2.2) is performed in due time, the European patent application will be deemed to be withdrawn (Rule 71(7) EPC).

4. Further procedure

4.1 In the case of point 2.1 above

4.1.1 The decision to grant the European patent will be issued, and the **mention of the grant** of the patent will be published in the European Patent Bulletin, if the requirements concerning the translation of the claims and the payment of all fees are fulfilled and there is agreement as to the text to be granted (Rule 71a(1) EPC).

Note on payment of the renewal fee:
If a renewal fee becomes due before the next possible date for publication of the mention of the grant of the European patent, publication will be effected only after the renewal fee and any additional fee have been paid (Rule 71a(4) EPC).

Under Article 86(2) EPC, the obligation to pay renewal fees to the European Patent Office terminates with the payment of the renewal fee due in respect of the year in which the mention of the grant of the European patent is published.

Note on payment of the designation fee(s):
If the designation fee(s) become(s) due after the communication under Rule 71(3) EPC, the mention of the grant of the European patent will not be published until these fees have been paid (Rule 71a(3) EPC).

4.1.2 After publication, the **European patent specification** can be downloaded free of charge from the EPO publication server https://data.epo.org/publication-server.

4.1.3 Filing of translations in the contracting states

As regards translation requirements prescribed by the contracting states under Article 65(1) EPC, please consult the website of the European Patent Office
www.epo.org ➞Law & practice ➞Legal texts, National law relating to the EPC
www.epo.org ➞Law & practice ➞All Legal texts ➞London Agreement

In the case of a valid extension or validation

As regards translation requirements prescribed by the extension or validation states, please consult the website of the European Patent Office
www.epo.org ➞Law & practice ➞Legal texts, National law relating to the EPC

Failure to supply a prescribed translation in a contracting state, or in an extension or validation state may result in the patent being deemed to be void *ab initio* in the state concerned (Art. 65(3) EPC).

4.2 In the case of 2.2 above

If the present communication under Rule 71(3) EPC is based on an auxiliary request and, within the time limit, you maintain the main request or a higher ranking request which is not allowable, the application will be refused (Art. 97(2) EPC).

If the examining division gives its consent to the requested amendments or corrections, it will issue a new communication under Rule 71(3) EPC; otherwise, it shall resume the examination proceedings (Rule 71(6) EPC).

5. Filing of a divisional application

Any divisional application relating to this European patent application must be filed directly with the European Patent Office in Munich, The Hague or Berlin and will be in the language of the proceedings for the present application, or if the latter was not in an official language of the EPO, the divisional application may be filed in the language of the present application as filed (see Article 76(1) and Rule 36(2) EPC). Any such divisional application must be filed while the present application is still pending (Rule 36(1) EPC; Guidelines A-IV, 1.1.1).

6. Notes concerning fee payments

6.1 Making payments

For payments made via deposit account, please note that as from 1 December 2017 debit orders will only be carried out if filed in an electronically processable format (xml), using an accepted means of filing as laid down in the Arrangements for deposit accounts (ADA), published in the Supplementary publication in the Official Journal.

All relevant information related to the modes of payment of fees to the EPO can be retrieved from the EPO website at "**Making Payments**".

6.2 Information concerning fee amounts

Procedural fees are usually adjusted every two years, on even years, with effect from 1 April. Therefore, before making a payment, parties should verify the amounts actually due on the date of payment using the applicable version of the Schedule of fees and expenses, published as a Supplement to the Official Journal of the EPO, available on the EPO website (**www.epo.org**) at **www.epo.org/schedule-of-fees**. The "Schedule of fees" table allows the viewing, downloading and searching of individual fee amounts, both current and previous.

6.3 Note to users of the automatic debiting procedure

The fee for grant, including the fee for publication, and any additional claims fees due under Rule 71(4) EPC will be debited automatically on the date of filing of the translations of the claims, or on the last day of the period of this communication. However, if the designation fee(s) become(s) due as set out in Rule 71a(3) EPC and/or a renewal fee becomes due as set out in Rule 71a(4) EPC, these should be paid separately by another permitted way of payment in order not to delay the publication of the mention of the grant. The same applies in these circumstances to the payment of extension and validation fees.

Examining Division:

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Enclosures: Text intended for grant

EPO Form 2056
EPO Form 2906

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Annex to EPO Form 2004, Communication pursuant to Rule 71(3) EPC

Bibliographical data of European patent application No. 20 794 947.0

For the intended grant of the European patent, the bibliographical data are set out below, for information:

Title of invention:
- RADON-SENSORVORRICHTUNG MIT POLYEDRISCHER IONISATIONSKAMMER
- RADON SENSOR DEVICE USING POLYHEDRAL-SHAPED IONIZATION CHAMBER
- DISPOSITIF DE CAPTEUR DE RADON UTILISANT UNE CHAMBRE D'IONISATION DE FORME POLYÉDRIQUE

Classification: INV. G01T1/185 G01T1/178 G01T7/00

Date of filing: 17.01.2020

Priority claimed: KR / 24.04.2019 / KRA20190047674

Contracting States* for which fees have been paid: AL AT BE BG CH CY CZ DE DK EE ES FI FR GB GR HR HU IE IS IT LI LT LU LV MC MK MT NL NO PL PT RO RS SE SI SK SM TR

Extension States* for which fees have been paid:

Validation States* for which fees have been paid:

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*) If the time limit for the payment of designation fees according to Rule 39(1) EPC has not yet

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expired and the applicant has not withdrawn any designation, **all Contracting States/Extension States/Validation States** are currently still deemed to be designated.
See also Rule 71a(3) EPC and, if applicable, the above Note to users of the automatic debiting procedure.

**) If two or more applicants have designated different Contracting States, this is indicated here.

National prior rights within the meaning of Article 139(2) EPC are not a bar to the grant of a European patent in proceedings before the EPO. Therefore, the EPO is not required to search for and assess such rights (see GL, H-III, 4.4). Applicants may, however, consider the procedural option under Rule 138 EPC in view of the effects of such rights in national proceedings and/or before the Unified Patent Court (Article 3 Regulation (EU) No 1257/2012). As a support service free of charge for the applicant in this context, the applicant is hereby offered non-binding information on a search for and prima facie relevance assessment of national prior rights by the examining division. It is the applicant's responsibility to assess such national prior rights and any use of the procedural option under Rule 138 EPC (see GL, H-III, 4.4).

The applicant is informed that no prima facie relevant national prior rights were found.

DESCRIPTION

RADON SENSOR DEVICE USING POLYHEDRAL-SHAPED IONIZATION CHAMBER

Technical Field

[0001] The present disclosure relates to a radon sensor using an ionization chamber and, more particularly, to a radon sensor device using a polyhedral-shaped ionization chamber.

Background Art

[0002] In general, an ionization chamber that is used for a radon sensor, as shown in FIG. 1, has a cylindrical structure in which a conductive cylinder having a plurality of pores and a straight probe unit disposed in the central axis in the cylinder are combined.

[0003] Such a cylindrical structure of the related art has the advantage that an electrical field intensifies toward the probe unit from the inner surface of the ionization chamber and is uniformly distributed in the axial direction, so the measurement efficiency is high. However, the cylindrical structure is difficult to mount and is vulnerable to electromagnetic wave noise due to the pores.

[0004] Further, there is inconvenience in that it is required to attach an appropriate filter to prevent dust from permeating through the pores. Further, since high-voltage bias is applied to the ionization chamber having a cylindrical structure and is completely exposed to the outside, there is also a possibility of leakage of electricity and an electric shock.

[0005] Further, air excessively smoothly flows inside due to the pores and thoron (Rn-220) having a short half-life of 55.6 seconds also flows into the ionization chamber, so there is a problem that there is a possibility of simultaneous measurement of radon (Rn-222) and thoron (Rn-220). WO9113371A1, US4853536A1,

JP2005156463A and KR1020170023601A are examples of radon sensors.

Disclosure

Technical Problem

[0006] The present disclosure has been made in an effort to solve the problems described above and an objective of the present disclosure is to provide a radon sensor device using a polyhedral-shaped ionization chamber that has excellent mountability and mass productivity, resists well electromagnetic noise, has not possibility of leakage of electricity, has high measurement efficiency, and can exclude interference by thoron (Rn-220) instead of the cylindrical structure of the related art.

Technical Solution

[0007] In order to achieve the objectives, the present invention provides a radon sensor device using a polyhedral-shaped ionization chamber 2 according to ~~and including the further features as claimed in~~ claim 1. The radon sensor device includes: an ionization chamber having an open side and inner sides surrounded by a first conductor and generating an electrical field therein by applying bias power to the first conductor; a cover having a first side covered with a second conductor and closing the open side of the ionization chamber such that that first conductor disposed on the inner sides of the ionization chamber and the second conductor are electrically connected; a probe unit disposed in the ionization chamber and absorbing ion charges produced when alpha (α) decay occurs in the ionization chamber; and a measurement circuit detecting an alpha particle detection signal by amplifying and processing an electrical micro-signal input from the probe unit into a predetermined magnitude.

[0008] The ionization chamber may be formed in a hexahedral shape using an insulating material.

[0009] The first conductor disposed on the inner sides of the ionization chamber may be formed by coating conductive paint with a predetermined thickness on the inner sides of the ionization chamber.

[0010] The bias power that is applied to the first conductor disposed on the inner sides of the ionization chamber is a DC voltage of 50V to 300V.

[0011] The first conductor disposed on the inner sides of the ionization chamber and the second conductor disposed on the first side of the cover may be electrically connected through joints between the cover and the ionization chamber.

[0012] The cover may be a printed circuit board (PCB) having a predetermined thickness and the entire of a first side of the printed circuit board may be covered with the second conductor.

[0013] A plurality of holes may be formed through the cover so that external air can flow into the ionization chamber or air can flow outside.

[0014] When a volume of the ionization chamber is 100cc, a diameter of the holes formed through the cover may be 0.2mm to 0.3mm and the number thereof may be 6 to 10.

[0015] The probe may detect ion charges produced by a collision of air and alpha particles produced when alpha (α) decay occurs in the ionization chamber.

[0016] The probe unit is disposed at a center in the ionization chamber and is formed in a T-shape.

[0017] The probe unit is formed in a T-shape by perpendicularly combining first and second probes that have a rod shape or a straight shape having a predetermined length; and a first end of the first probe is electrically connected with the measurement circuit through the cover and a second end of the first probe is disposed at the center in the ionization chamber, and a center of the second probe is perpendicularly coupled to the second end of the first probe and is disposed at the center in the ionization chamber such that distances between the inner sides of the ionization chamber and a surrounding portion of the second probe are the same.

[0018] The probe unit is disposed at a center in the ionization chamber. ~~chamber and is formed in an outdoor TV antenna shape.~~

[0019] According to an example useful for understanding, but not forming part of, the presently claimed invention, ~~embodiment as disclosed,~~ the probe unit is formed by combining a first probe having a T-shape and second probes having a predetermined

pattern, in which the first probe has an upper head formed in a rod shape or a straight shape with a predetermined length and disposed at the center in the ionization chamber, has a lower leg formed in a rod shape or a straight shape with a predetermined length and having a first end perpendicularly connected to a center of the upper head to form a T-shape and a second end electrically connected to the measurement circuit through the cover; and the second probes are formed in a rod shape or a straight shape with a predetermined length, are coupled at centers to the upper head of the first probe, are longitudinally spaced a predetermined distance from each other, and are coupled perpendicular to or at a predetermined angle with respect to a longitudinal direction of the upper head of the first probe to have a predetermined pattern.

[0020] According to the present invention, the probe unit is formed by combining a first probe having a T-shape and second probes having a predetermined pattern, in which the first probe has an upper head formed in a rod shape or a straight shape with a predetermined length and disposed at the center in the ionization chamber, has a lower leg formed in a rod shape or a straight shape with a predetermined length and having a first end perpendicularly connected to a center of the upper head to form a T-shape and a second end electrically connected to the measurement circuit through the cover; and the second probes are combined to vertically and horizontally cross each other at a right angle in a lattice pattern and centers thereof are longitudinally coupled to the upper head of the first probe.

[0021] The second probes may be coupled to the upper head of the first probe to vertically and horizontally cross each other at a right angle, whereby the lattice pattern thereof may be a square lattice pattern or a skew-crossed pattern.

[0022] The probe unit may be disposed in the ionization chamber such that distances between the inner sides of the ionization chamber and a surrounding portion of the second probe are the same.

[0023] The measurement circuit may be disposed on a second side of the cover and may be mounted to be electrically connected with the second conductor

disposed on the first side of the cover through a bias voltage outputter for outputting the bias power.

[0024] The measurement circuit may include: a guard ring installed such that the probe unit is disposed therethrough and absorbing and sensing a leakage current generated between the ionization chamber and the probe unit to a ground; a noise detector connected between the guard ring and the ground, sends a DC-type leakage current generated between the ionization chamber and the probe to the ground, and detects and outputs AC-type noise; first and second pre-amplifiers respectively connected to another end of the probe and an output end of the noise detector, and amplifying small electrical signals input from the probe and the noise detector into a predetermined magnitude; and a differential amplifier having a non-inverting terminal (+) and an inverting terminal (-) respectively connected to output ends of the first and second pre-amplifiers, and offsetting noise signals and outputting an alpha particle detection signal by amplifying a voltage difference of electrical signals pre-amplified through the first and second pre-amplifiers.

[0025] The noise detector may be configured by electrically connecting at least any one element of a resistance, a condenser, and a diode in series, in parallel, or in combination of series and parallel.

[0026] The radon sensor device may further include an electromagnetic wave-protective cap installed on a second side of the cover to surround the entire measurement circuit and is made of a conductive material to prevent signal disturbance of the measurement circuit due to external electromagnetic wave noise.

[0027] The electromagnetic wave-protective cap may be installed to be electrically connected with a ground of the measurement circuit.

Advantageous Effects

[0028] As described above, according to the radon sensor device using a polyhedral-shaped ionization chamber, it is possible to provide a radon sensor that has excellent mountability, resists well electromagnetic wave noise, is simply assembled

with a low cost, and is suitable for mass production by using a polyhedral-shaped ionization chamber that has excellent mountability and mass productivity, resists well electromagnetic wave noise, has high measurement efficiency without a possibility of an electric shock or leakage of electricity, and can exclude interference of thoron (Rn-220) instead of an ionization chamber having a cylindrical shape of the related art.

Description of Drawings

[0029] FIG. 1 is a perspective view showing an ionization chamber having a cylindrical structure that is applied to the related art;

[0030] FIG. 2 is an assembled perspective view showing a radon sensor device using a polyhedral-shaped ionization chamber according to an example useful for understanding, but not forming part of, the presently claimed invention. ~~embodiment of the present disclosure;~~

[0031] FIG. 3 is an assembled cross-sectional view showing the radon sensor device using a polyhedral-shaped ionization chamber according to an example useful for understanding, but not forming part of, the presently claimed invention. ~~embodiment of the present disclosure;~~

[0032] FIGS. 4a to 4d are perspective view showing a probe unit applied to another embodiment of the present invention; ~~disclosure;~~ and

[0033] FIG. 5 is a detailed block diagram showing a measurement circuit applied to an embodiment of the present invention. ~~disclosure.~~

Best Mode

[0034] The objectives, characteristics, and advantages will be described in detail below with reference to the accompanying drawings, so those skilled in the art may easily achieve the scope of the present invention. ~~disclosure.~~ In describing the present invention, ~~disclosure,~~ detailed descriptions of well-known technologies will be omitted so as not to obscure the description of the present invention ~~disclosure~~ with unnecessary details.

[0035] Terms including ordinal numbers such as ‘first’, ‘second’, etc., may be used to describe various components, but the components are not to be construed as being limited to the terms. The terms are used only to distinguish one component

from another component. For example, the 'first' component may be named the 'second' component, and vice versa, without departing from the scope of the present invention.

~~disclosure.~~ Terms used in the present specification are used only to describe specific exemplary embodiments rather than limiting the present invention. ~~disclosure.~~ Singular forms are intended to include plural forms unless the context clearly indicates otherwise.

[0036] Terminologies used herein were selected as terminologies that are currently used as generally as possible in consideration of the functions herein, but may be changed, depending on the intention of those skilled in the art, precedents, or advent of a new technology. Further, there are terminologies selected by applicant(s) at the applicant(s)' opinion in specific cases, and in these cases, the meanings will be described in the corresponding parts. Accordingly, the terminologies used herein should be defined on the basis of the meanings of the terminologies and the entire specification, not simply the names of the terminologies.

[0037] Throughout the present specification, unless explicitly described otherwise, "comprising" any components will be understood to imply the inclusion of other components rather than the exclusion of any other components. Further, the terms "~ part", "module", and the like mean a unit for processing at least one function or operation and may be implemented by hardware or software or by a combination of hardware and software.

[0038] Hereinafter, embodiments of the present invention ~~disclosure~~ will be described in detail with reference to the accompanying drawings. However, the embodiments of the present invention ~~disclosure~~ that will be described hereafter may be modified in various ways and the scope of the present invention ~~disclosure~~ is not limited to the embodiments that will be described in detail below. Embodiments of the present invention ~~disclosure~~ are provided to more completely describe the present invention ~~disclosure~~ to those skilled in the art.

[0039] FIG. 2 is an assembled perspective view showing a radon sensor device using a polyhedral-shaped ionization chamber according to an example useful for understanding, but not forming part of, the presently claimed invention, ~~embodiment of~~

~~the present disclosure;~~ FIG. 3 is an assembled cross-sectional view showing the radon sensor device using a polyhedral-shaped ionization chamber according to an

example useful for understanding, but not forming part of, the presently claimed invention, ~~embodiment of the present disclosure,~~ FIGS. 4a to 4d are perspective views showing a probe unit applied to another embodiment of the present invention, ~~disclosure,~~ and FIG. 5 is a detailed block diagram showing a measurement circuit applied to an embodiment of the present invention. ~~disclosure.~~

[0040] Referring to FIGS. 2 to 5, a radon sensor device using a polyhedral-shaped ionization chamber, ~~chamber according to an embodiment of the present disclosure, in a broad meaning,~~ includes an ionization chamber 100, cover 200, a support 300, a measurement circuit 400, etc. The polyhedral-shaped ionization chamber according to an embodiment of the present invention ~~disclosure~~ may further include an electromagnetic wave-protective cap 500, etc. ~~The components shown in FIGS. 2 to 5 are not necessary, so the radon sensor device using a polyhedral-shaped ionization chamber according to an embodiment of the present disclosure may include more components or fewer components.~~

[0041] Hereafter, the components of the radon sensor device using a polyhedral-shaped ionization chamber according to the invention ~~an embodiment of the present disclosure~~ are described in detail.

[0042] The ionization chamber 100 has a polyhedral shape having an open side and inner sides surrounded by a first conductor 110 made of a conductive material, and generates an electrical field therein by applying bias power 10 to the first conductor 110.

[0043] It is preferable that the ionization chamber 100 is formed entirely in a hexahedral shape (e.g., a regular hexahedron or a rectangular parallelepiped) using an insulating material such as plastic, but is not limited thereto and may be a solid figure surrounded with four or more flat polygons such as a tetrahedron, a pentahedron, a heptahedron, and an octahedron.

[0044] Further, it is preferable that the first conductor 110 disposed on the inner sides of the ionization chamber 100 may be formed by coating conductive paint to have

a predetermined thickness on the inner sides of the ionization chamber 100.

[0045] Further, it is preferable that the bias power 10 that is applied to the first

conductor 110 disposed on the inner sides of the ionization chamber 100 may be a DC voltage of about 50V to 300V.

[0046] An electrical field is generated between the probe unit 300 in the ionization chamber 100, the first conductor 110 disposed on the inner sides of the ionization chamber 100, and a second conductor 210 disposed on a side of the cover 200 by applying high-voltage bias power 10 to the first conductor 110 disposed on the inner sides of the ionization chamber 100 and the second conductor 210 disposed on a side of the cover 200 to be described below, whereby it is possible to absorb an ion current generated by alpha (α) decay through the probe unit 300. The ionization chamber 100 has a simple configuration, can be implemented at a low cost, and can perform 3-dimensional measurement.

[0047] Further, a stable DC voltage of about 50V to 300V (more preferably, about 50V to 150V) is used for the high-voltage bias power 10 that is applied to the first conductor 110 disposed on the inner sides of the ionization chamber 100 and the second conductor 210 disposed on a side of the cover 200, thereby obtaining a condition in which additional ion charges can be effectively produced when alpha decay occurs in the ionization chamber 100. The power supplied from the high-voltage bias power 10 (e.g., a voltage or a current) may be changed in various ways, depending on the measurement range, the sensitivity, etc.

[0048] The cover 200 closes the open side of the ionization chamber 100, has a side covered with the second conductor 210 made of a conductive material, and electrically connects the first conductor 110 disposed on the inner sides of the ionization chamber 100 and the second conductor 210, and is coupled such that the side on which the second conductor 210 is disposed and the open side of the ionization chamber 100 face each other, thereby closing the open side of the ionization chamber 100.

[0049] In this configuration, it is preferable that the first conductor 110 disposed on the inner sides of the ionization chamber 100 and the second conductor 210 disposed on a side of the cover 200 are electrically connected through the joints C

between the cover 200 and the ionization chamber 100.

[0050] The cover 200 may be a common printed circuit board (PCB) having a predetermined thickness and one entire side of the printed circuit board (PCB) may be covered with the second conductor 210.

[0051] Further, a plurality of 220 may be formed through the cover 200 so that external air can flow into the ionization chamber 100 or air can flow outside.

[0052] The diameter and the number of the holes 220 formed through the cover 200 may depend on the volume of the ionization chamber 100.

[0053] For example, when the volume of the ionization chamber is 100cc, it is preferable that the diameter is about 0.2mm to 0.3mm and the number is about 6 or 10.

[0054] In this case, since a diffusion time for external air to flow into the ionization chamber 100 is about 5 minutes to 10 minutes, the half-life of thoron (Rn-220) which is about 55.6 seconds is repeated about 5 times or more, so the concentration decreases to $1/2^5$ or less. Accordingly, interference by thoron (Rn-220) is excluded.

[0055] Further, since the holes 220 are small and large dust cannot flow inside, there is no need for a separate dust filter. Further, the open space of the ionization chamber 100 is minimized and the possibility to cause a measurement error due to permeation of electromagnetic wave noise is remarkably decreased in comparison to the ionization chamber having a cylindrical structure with several pores shown in FIG. 1.

[0056] Meanwhile, when the cover 200 is a printed circuit board (PCB), it is preferable that the holes 220 are pinholes of small defects or small holes formed through a film such as a metallic deposited film or an insulating film.

[0057] The probe unit 300 is made of a conductive material to be able to absorb ion charges produced when alpha decay occurs due to a radon (Rn-222) nuclide, etc. in the ionization chamber, is disposed in the ionization chamber 100, and absorbs ion charges produced when alpha (α) decay occurs in the ionization chamber

100.

[0058] That is, the probe unit 300 may be provided to detect ion charges produced by a collision of air and alpha particles produced when alpha (α) decay occurs in the ionization chamber 100.

[0059] It is preferable that the probe unit 300 is provided to detect alpha particles produced when alpha decay occurs due to radon (Rn-222) gas from the air flowing in the ionization chamber 100, but it is not limited thereto and may be provided to detect all radioactive gases discharged as alpha (α) particles.

[0060] Further, the probe unit 300 is formed in a T-shape, as shown in FIGS. 2 and 3 as 2 examples useful for understanding, but not forming part of, the presently claimed invention, 2 ~~an embodiment,~~ so it is preferable that the probe unit 300 is disposed at the center in the ionization chamber 100.

[0061] That is, the probe unit 300 is formed in a T-shape by perpendicularly combine a first probe 300a and a second probe 300b that is formed in a rod shape or a straight shape with a predetermined length. A first end of the first probe 300a is electrically connected to the measurement circuit 400 through the cover 200 and a second end is disposed at the center in the ionization chamber 100.

[0062] Further, the center of the second probe 300b is perpendicularly coupled to the second end of the first probe 300a and is disposed at the center in the ionization chamber 100 such that the distances between the inner sides of the ionization chamber 100 and the surrounding portion of the second probe 300b are the same.

[0063] Further, the probe unit 300', as shown in FIGS. 4a to 4d as 2 embodiments of the invention, 2 ~~other embodiments,~~ may be formed in the shape of a common outdoor TV antenna and is preferably disposed at the center in the ionization chamber 100.

[0064] That is, as shown in FIGS. 4a and 4b, the probe unit 300' is formed by combining a T-shaped first probe 300a' and second probes 300b' having a predetermined pattern, and the first probe 300a' may be formed in the same shape as the probe unit 300 described above. That is, the first probe 300a' has an upper head formed in a rod shape or a straight shape with a predetermined length and disposed at the center in the ionization chamber 100, has a lower leg formed in a rod shape or

a straight shape with a predetermined length and having a first end perpendicularly connected to the center of the upper head to form a T-shape and a second end electrically connected to the measurement circuit 400 through the cover 200.

[0065] Further, the second probes 300b' are formed in a rod shape or a straight shape with a predetermined length, are coupled at the centers to the upper head of the first probe 300a', and are longitudinally spaced a predetermined distance from each other. Further, the second probes 300b' may be coupled perpendicular to (see FIG. 4a) or at a predetermined angle θ (preferably, about 1 to 90 degrees) (FIG. 4b) with respect to the longitudinal direction of the upper head of the first probe 300a' in the same plane as the upper head of the first probe 300a', thereby having a predetermined pattern.

[0066] Further, as shown in FIGS. 4c and 4d, the probe unit 300' is formed by combining a T-shaped first probe 300a' and second probes 300b' having a predetermined pattern. The upper head of the first probe 300a' is formed in a rod shape or a straight shape with a predetermined length and is disposed at the center in the ionization chamber 100. The lower leg of the first probe 300a' is formed in a rod shape or a straight shape with a predetermined length and has a first end perpendicularly coupled to the center of the upper head and a second end electrically connected to the measurement circuit 400 through the cover 200.

[0067] Further, the second probes 300b' are combined to vertically and horizontally cross each other at a right angle in a lattice pattern and the centers thereof may be longitudinally coupled to the upper head of the first probe 300a'.

[0068] Further, the second probes 300b', as shown in FIG. 4c, are coupled to the upper head of the first probe 300a' to vertically and horizontally cross each other at a right angle in the same plane as the upper head of the first probe 300a', whereby the lattice pattern thereof may be a square lattice pattern. Further, as shown in FIG. 4d, the lattice pattern may be a skew-crossed pattern. Further, though not shown in the figures, for example, the lattice pattern may be formed in various shapes such as a rectangular lattice pattern, a square lattice pattern, or a changed lattice pattern.

[0069] That is, it is advantageous that the probe unit 300 or 300' is a T-shape or an outdoor TV antenna shape rather than a straight probe that is generally used in the ionization chamber having a cylindrical shape shown in FIG. 1 described above. The reason is because when a straight shape is applied to a polyhedral ionization chamber 100, the distance between the inner sides and the probe 300 or 300' is large and is not uniform, so it is difficult to maintain an intense electrical field in the ionization chamber 100.

[0070] Accordingly, the probe unit 300 or 300' should be designed such that the distance between the inner sides of the ionization chamber 100 and the probe unit 300 or 300' is maintained as small as possible and an intense electrical field can be generated around the probe unit 300 or 300'. When the probe unit 300 or 300' having a T-shape or an outdoor TV antenna shape proposed in embodiments of the present invention ~~disclosure~~ are designed, the distance from the inner sides of the ionization chamber 100 is small and the intensity of the electrical field around the probe unit 300 or 300' is also increased, whereby absorption efficiency when ion charges are produced is increased and accordingly the measurement efficiency can be increased.

[0071] That is, the probe unit 300 or 300' is designed to be disposed in the ionization chamber 100 such that the distances between the inner sides of the ionization chamber 100 and the surrounding portion of the probes are the same, whereby the distances between the surrounding portion of the probe unit 300 or 300' and the inner sides of the ionization chamber 100 are uniform and small and the intensity of the electrical field around the probe unit 300 or 300' is also increased. Accordingly, it is possible to effectively increase the measurement efficiency by increasing the absorption efficiency when ion charges are produced.

[0072] The probe unit 300 or 300' configured in this way is electrically connected to the measurement circuit 400 through the cover 200.

[0073] Further, the measurement circuit 400 outputs an alpha particle detection signal by amplifying and processing an electrical micro-signal input from the probe unit 300 or 300' into a predetermined magnitude.

[0074] It is preferable that the measurement circuit 400 is disposed on a second side of the cover 200 and is mounted to be electrically connected with the second conductor 210 disposed on a first side of the cover 200 through a bias voltage outputter 401 for outputting bias power 10.

[0075] Further, the measurement circuit 400, as shown in FIG. 5, in a broad meaning, may include a guard ring 410, a noise detector 420, a first pre-amplifier 430, a second pre-amplifier 440, and/or a differential amplifier 450.

[0076] The guard ring 410 is made of a conductive material in a cylindrical shape, has the probe unit 300 or 300' disposed therethrough, and absorbs and sends a leakage current generated between the ionization chamber 100 and the probe unit 300 or 300' to a ground.

[0077] If the guard ring 410 is not provided, a current signal obtained from the probe unit 300 or 300' and a leakage current signal are combined, which causes the problem that the signal-to-noise ratio (SNR) decreases.

[0078] The noise detector 420 is connected between the guard ring 410 and the ground and maintains the potential of the guard ring 410 a little bit higher than the ground, thereby sending a DC-type leakage current generated between the ionization chamber 100 and the probe unit 300 or 300' to the ground and detecting and outputting an AC-type noise to the second pre-amplifier 440 to be described below.

[0079] It is preferable to configure the noise detector 420, for example, by electrically connecting at least any one element of a resistance, a condenser, and a diode in series, in parallel, and/or in combination of series and parallel.

[0080] The first pre-amplifier 430 is connected to a second end of the probe unit 300 or 300' and amplifies an electrical micro-signal input from the probe unit 300 or 300' into a predetermined magnitude.

[0081] The second pre-amplifier 440 is connected to an output end of the noise detector 420 and amplifies an electrical micro-signal input from the noise detector 420 into a predetermined magnitude.

[0082] Output ends of the first pre-amplifier 430 and the second pre-amplifier

440 are respectively connected to a non-inverting terminal (+) and an inverting terminal (-) of the differential amplifier 450. The differential amplifier 450 can effectively offset noise signals and can detect an alpha particle detection signal by amplifying the electrical signals pre-amplified through the first pre-amplifier 430 and the second pre-amplifier 440 in proportion to the voltage difference of the electrical signals.

[0083] That is, as differential amplification is performed by the differential amplifier 450, it is possible to effectively offset noises having the same phase and flowing inside through the probe unit 300 or 300' and the noise detector 420 and it is possible to accurately and quickly obtain a high-sensitivity and low-noise alpha particle detection signal.

[0084] Further, the electromagnetic wave-protective cap 500 is installed on the second side of the cover 200 to surround the entire measurement circuit 400 and is made of a conductive material to prevent signal disturbance of the measurement circuit 400 due to external electromagnetic wave noise.

[0085] It is preferable that the electromagnetic wave-protective cap 500 is installed to be electrically connected with the ground of the measurement circuit 400.

[0086] Meanwhile, though not shown in the figures, a control unit (e.g., an MCU) that calculates a concentration value of alpha particles (e.g., a radon concentration value) by counting alpha particle detection signals (e.g., pulse signals), which are output from the differential amplifier 450, for a predetermined measurement time (e.g., 10 minutes, 30 minutes, and 1 hour), etc. may be provided.

[0087] That is, the present ㄹ invention ㄹ ~~disclosure~~ can effectively provide a radon sensor device using a polyhedral ionization chamber that has excellent mountability and mass productivity and can exclude interference of thoron (Rn-220).

[0088] Although preferred embodiments of a radon sensor device using a polyhedral ionization chamber according to the present ㄹ invention ㄹ ~~disclosure~~ were described above, the present ㄹ invention ㄹ ~~disclosure~~ is not limited thereto and may be modified in various ways without departing from claims, the detailed description, and the accompanying drawings, and they are also included in the present disclosure. ㄹ The scope of the invention is defined by the appended claims. ㄹ

Industrial Applicability

[0089] The present disclosure can be widely used in radon sensor devices.

CLAIMS

1. A radon sensor device using a polyhedral-shaped ionization chamber, the radon sensor device comprising:

said polyhedral-shaped ~~an~~ ionization chamber (100) having an open side and inner sides surrounded by a first conductor (110) and for generating an electrical field therein by applying bias power to the first conductor (110);

a cover (200) having a first side covered with a second conductor (210) and closing the open side of the ionization chamber (100) such that that first conductor (110) disposed on the inner sides of the ionization chamber (100) and the second conductor (210) are electrically connected;

a probe unit (300) disposed in the ionization chamber (100) and absorbing ion charges produced when alpha (α) decay occurs in the ionization chamber (100); and

a measurement circuit (400) detecting an alpha particle detection signal by amplifying and processing an electrical micro-signal input from the probe unit (300) into a predetermined magnitude, wherein ~~characterized in that~~ the probe unit (300) is disposed at a center in the ionization chamber (100)~~and is formed in an outdoor TV antenna shape~~,

characterized in that ~~wherein~~ the probe unit (300') is formed by combining a first probe (300a') having a T-shape and second probes (300b') having a predetermined pattern,

wherein the first probe (300a') has an upper head formed in a rod shape or a straight shape with a predetermined length and disposed at the center in the ionization chamber, has a lower leg formed in a rod shape or a straight shape with a predetermined length and having a first end perpendicularly connected to a center of the upper head to form a T-shape and a second end electrically connected to the measurement circuit (400) through the cover (200); and

the second probes (300b') are combined to vertically and horizontally cross each other at a right angle in a lattice pattern and centers thereof are longitudinally coupled to the upper head of the first probe.

2. The radon sensor device of claim 1, wherein the ionization chamber (100) is formed in a hexahedral shape using an insulating material.

3. The radon sensor device of claim 1, wherein the first conductor (110) disposed on the inner sides of the ionization chamber (100) is formed by coating conductive paint with a predetermined thickness on the inner sides of the ionization chamber (100).

4. The radon sensor device of claim 1, wherein the bias power that is applied to the first conductor (110) disposed on the inner sides of the ionization chamber is a DC voltage of 50V to 300V.

5. The radon sensor device of claim 1, wherein the first conductor (110) disposed on the inner sides of the ionization chamber (100) and the second conductor (210) disposed on the first side of the cover (200) are electrically connected through joints (C) between the cover (200) and the ionization chamber (100).

6. The radon sensor device of claim 1, wherein the cover (200) is a printed circuit board (PCB) having a predetermined thickness and the entire of a first side of the printed circuit board is covered with the second conductor (210).

7. The radon sensor device of claim 1, wherein a plurality of holes (220) is formed through the cover (200) so that external air can flow into the ionization chamber (100) or air can flow outside.

8. The radon sensor device of claim 7, wherein when a volume of the ionization chamber (100) is 100cc, a diameter of the holes (220) formed through the cover (200)

is 0.2mm to 0.3mm and the number thereof is 6 to 10.

9. The radon sensor device of claim 1, wherein the probe unit (300) detects ion charges produced by a collision of air and alpha particles produced when alpha (α) decay occurs in the ionization chamber (100).

~~10. The radon sensor device of claim 1, wherein the probe unit (300) is disposed at a center in the ionization chamber (100) and is formed in a T-shape.~~

~~11. The radon sensor device of claim 10, wherein the probe unit (300') is formed in a T-shape by perpendicularly combining first and second probes (300a', 300b') that have a rod shape or a straight shape having a predetermined length, and~~

~~a first end of the first probe (300a') is electrically connected with the measurement circuit (400) through the cover (200) and a second end of the first probe (300a') is disposed at the center in the ionization chamber (100), and a center of the second probe (300b') is perpendicularly coupled to the second end of the first probe (300a') and is disposed at the center in the ionization chamber (100) such that distances between the inner sides of the ionization chamber (100) and a surrounding portion of the second probe (300b') are the same.~~

[2] 10. [2]

~~12.~~ The radon sensor device of claim 1, wherein the second probes (300b') are coupled to the upper head of the first probe (300a') to vertically and horizontally cross each other at a right angle, whereby the lattice pattern thereof is a square lattice pattern or a skew-crossed pattern.

[2] 11. [2]

~~13.~~ The radon sensor device of claim 1, wherein the probe unit (300') is disposed in the ionization chamber (100) such that distances between the inner sides

of the ionization chamber (100) and a surrounding portion of the second probe (300b') are the same.

청구항 12.

~~14.~~ The radon sensor device of claim 1, wherein the measurement circuit (400) is disposed on a second side of the cover (200) and is mounted to be electrically connected with the second conductor (210) disposed on the first side of the cover (200) through a bias voltage outputter for outputting the bias power.

청구항 13.

~~15.~~ The radon sensor device of claim 1, wherein the measurement circuit (400) comprises:

a guard ring (410) installed such that the probe unit (300) is disposed therethrough and absorbing and sensing a leakage current generated between the ionization chamber (100) and the probe unit (300) to a ground;

a noise detector (420) connected between the guard ring(410) and the ground, sends a DC-type leakage current generated between the ionization chamber (100) and the probe (300) to the ground, and detects and outputs AC-type noise;

first and second pre-amplifiers (430, 440) respectively connected to another end of the probe (300) and an output end of the noise detector (420), and amplifying small electrical signals input from the probe (300) and the noise detector (420) into a predetermined magnitude; and

a differential amplifier (450) having a non-inverting terminal (+) and an inverting terminal (-) respectively connected to output ends of the first and second pre-amplifiers (430, 440), and offsetting noise signals and outputting an alpha particle detection signal by amplifying a voltage difference of electrical signals pre-amplified through the first and second pre-amplifiers (430, 440).

청구항 14.

~~16.~~ The radon sensor device of claim 청구항 13,

~~15,~~ wherein the noise detector (420) is

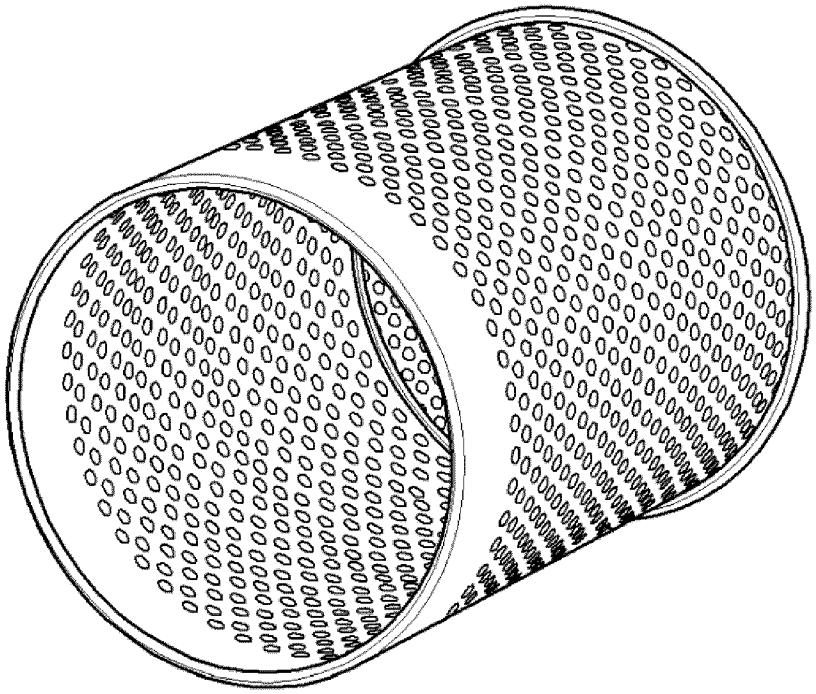
configured by electrically connecting at least any one element of a resistance, a condenser, and a diode in series, in parallel, or in combination of series and parallel.

2 15. 2 ~~17.~~ The radon sensor device of claim 1, further comprising;

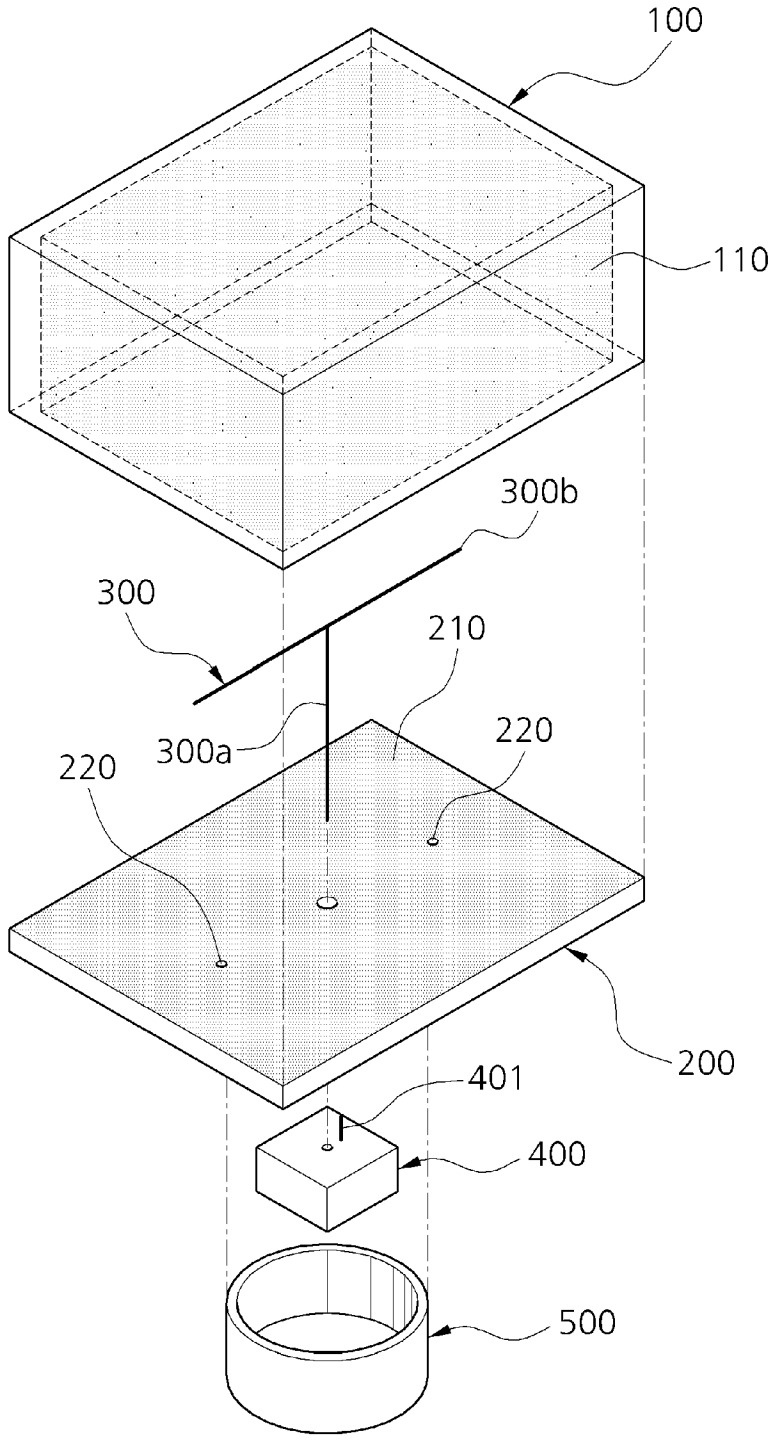
an electromagnetic wave-protective cap (500) installed on a second side of the cover (200) to surround the entire measurement circuit (400) and is made of a conductive material to prevent signal disturbance of the measurement circuit (400) due to external electromagnetic wave noise,

wherein the electromagnetic wave-protective cap (500) is installed to be electrically connected with a ground of the measurement circuit (400).

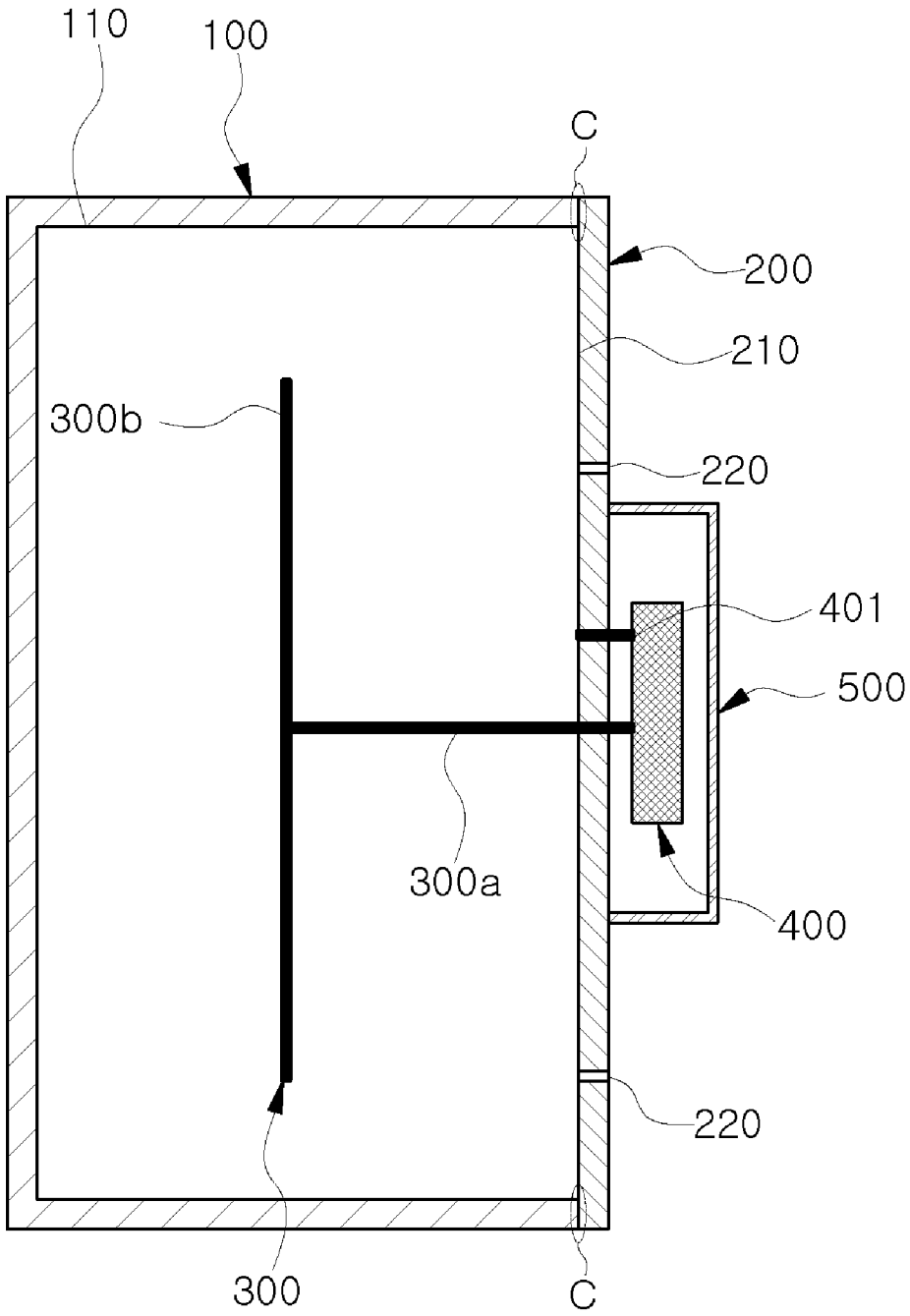
[FIG. 1]



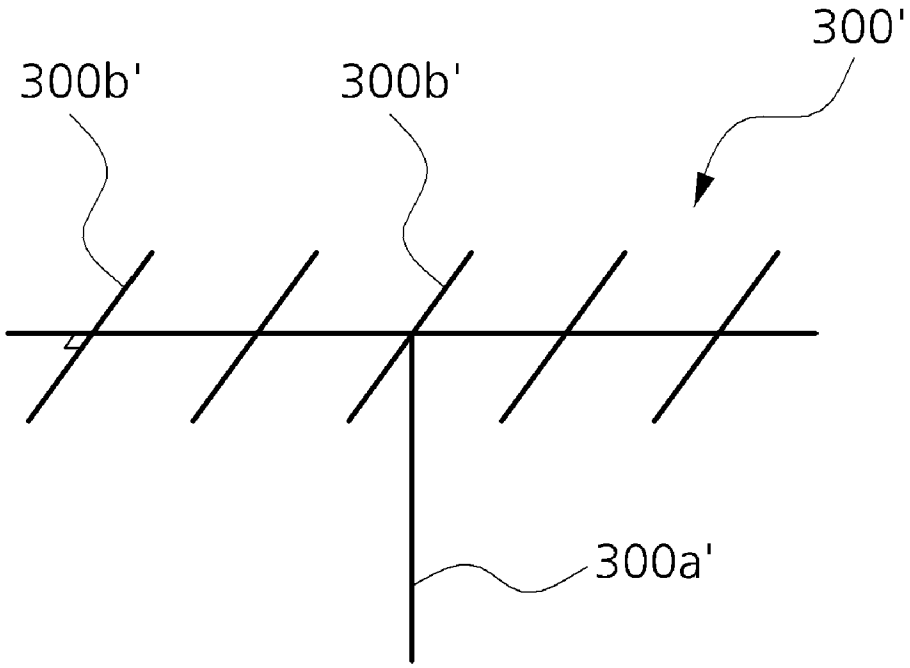
[FIG. 2]



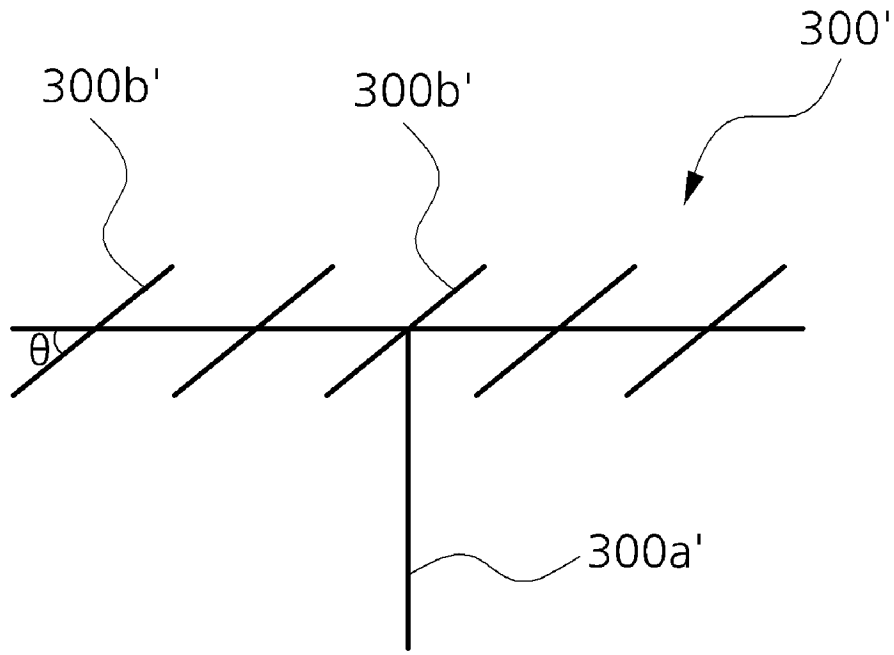
[FIG. 3]



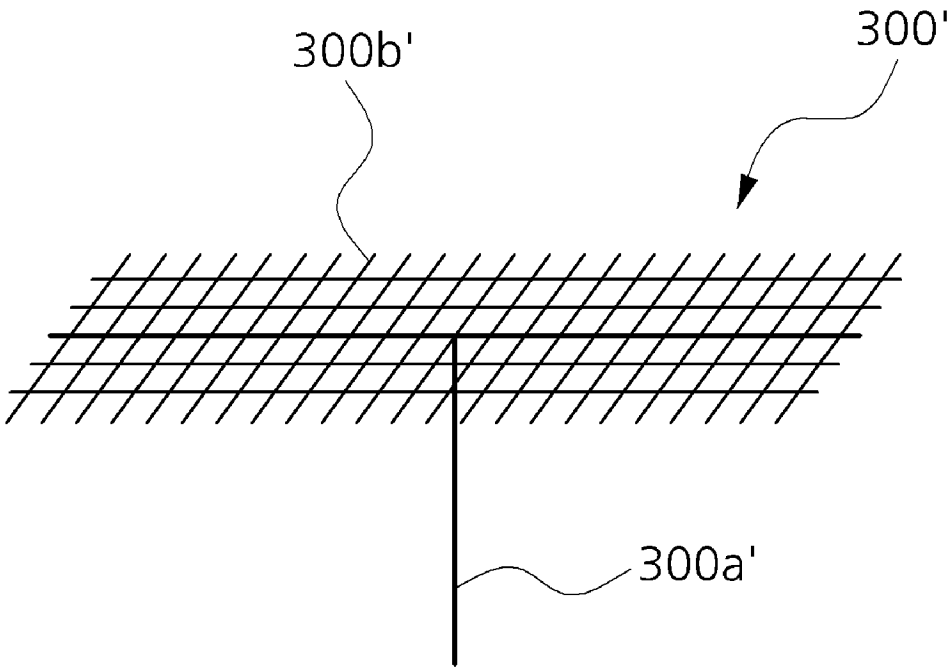
[FIG. 4A]



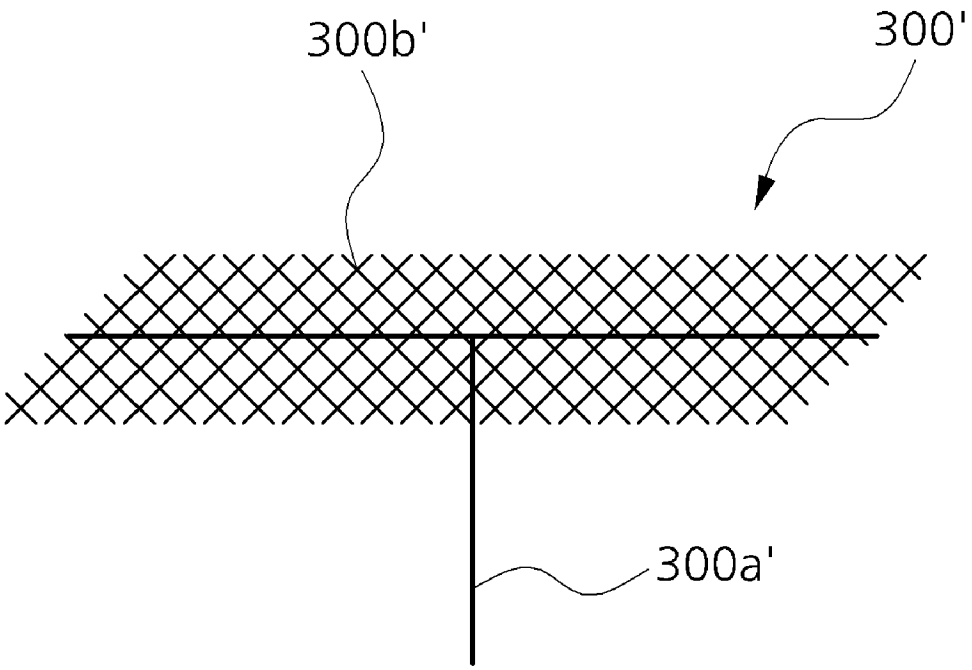
[FIG. 4B]



[FIG. 4C]

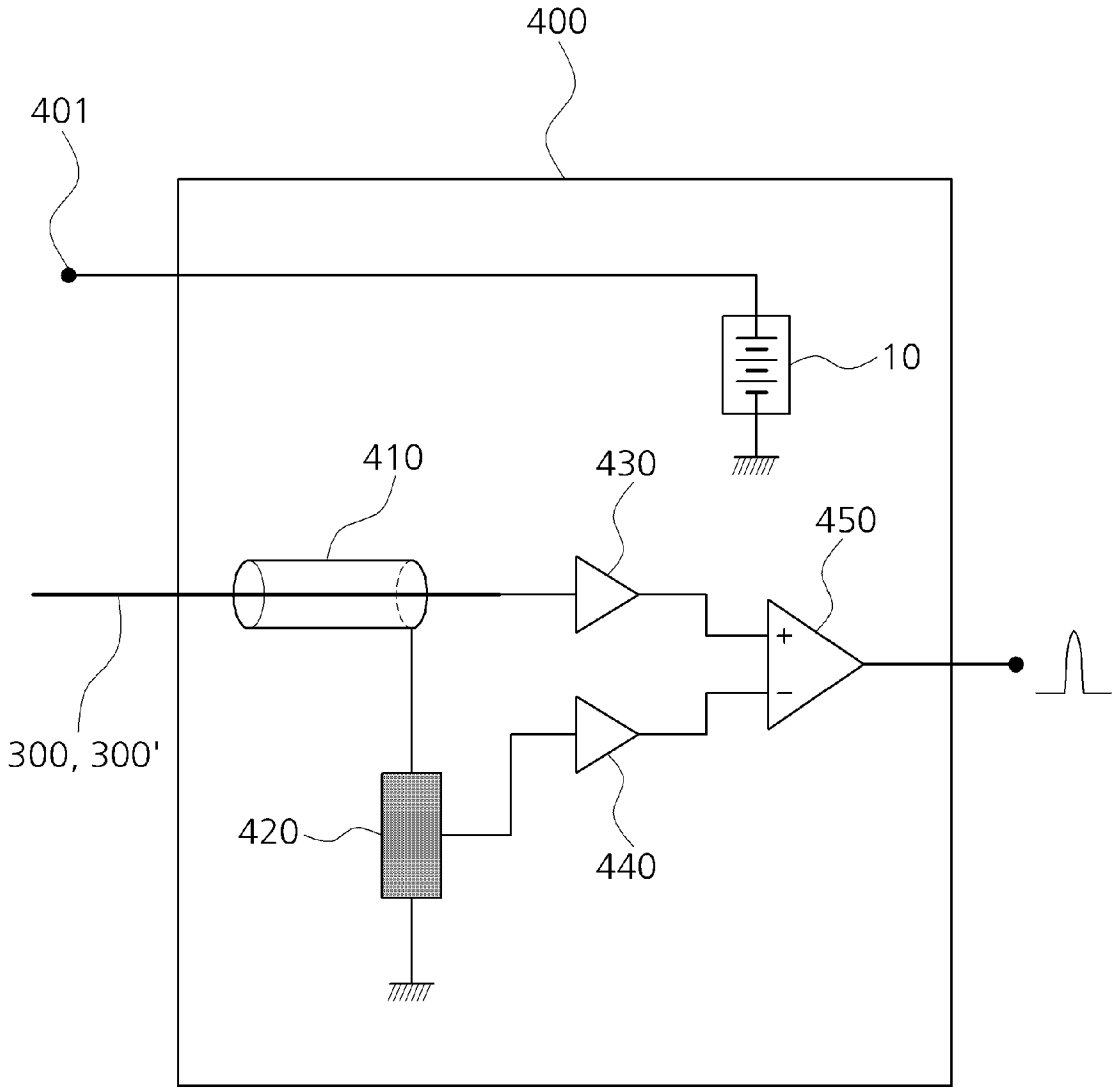


[FIG. 4D]



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[FIG. 5]



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